

Ticket 08 Study Report: Obstacle and Complementarity Diagnostics

SURF2026 American Option Risk Surfaces

Codex-assisted implementation report

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Abstract

Ticket 08 adds a diagnostic layer for American CN/PSOR results. It does not introduce a new pricing algorithm. Instead, it reconstructs the discrete linear complementarity problem solved at each American time step and checks the obstacle gap, equation gap, and complementarity product on interior spot nodes. The diagnostics are applied to the representative American put case from the validation ladder and the positive-dividend American call case from Ticket 07. This report is written for beginner researchers with accounting, data, machine-learning, and Python experience who are still learning obstacle problems, LCP diagnostics, and CN/PSOR validation.

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1 Role of Ticket 08 in the Whole Solver Project

Tickets 01 through 07 built the pricing and validation ladder: analytic Black-Scholes utilities, finite-difference grids, European Crank–Nicolson validation, American CN/PSOR solving, American put validation, no-dividend American call validation, and dividend-paying American call validation. Ticket 08 adds a diagnostic layer on top of those American solver outputs.

The key purpose is to check more than price levels. A numerical American option result can look reasonable and still fail the LCP logic. Ticket 08 therefore asks:

- Does the value stay above payoff?
- Does the reconstructed equation inequality hold?
- Is the product of the two LCP gaps close to zero?
- Did the PSOR iterations converge while these diagnostics were computed?

This ticket is not boundary extraction. It does not locate a free boundary, does not create Greeks, does not create stress maps, does not create datasets, and does not modify the solver.

2 Theory and Methodology/Literature Connection

The documented source basis for Ticket 08 is internal:

- `reports/01_literature/literature_map.md`;
- `reports/02_math/formulation_note.md`;
- `reports/03_solver/solver_validation_plan.md`;
- `reports/03_solver/solver_implementation_tickets.md`;
- the Ticket 01 through Ticket 07 implementation reports.

The formulation note explains that American exercise turns the pricing problem into a variational inequality and, after discretization, a linear complementarity problem. The solver validation plan explicitly calls for obstacle violation and complementarity residual reporting. The implementation ticket file says the diagnostics must keep boundary nodes separate from interior LCP residuals and must not hide maximum residuals behind averages.

This reference integration pass adds verified citations for the complementarity language used by the diagnostics. The American option discretization is interpreted through LCP methodology (Cottle et al., 2009) and the earlier finite-difference American option background (Brennan and Schwartz, 1977); the internal project documents remain the implementation source basis.

3 Obstacle, Equation Gap, and Complementarity in Plain Language

For an American option, the value U must be at least the payoff Φ :

$$U \geq \Phi. \tag{1}$$

This is the payoff obstacle. If a computed American value falls below payoff, the holder would be better off exercising immediately, so the numerical value is financially invalid.

The CN/PSOR solver also solves a continuation equation at every time step. In matrix form the interior-node LCP is:

$$Au \geq b, \quad u \geq \Phi, \quad (u - \Phi)^\top (Au - b) = 0. \quad (2)$$

The two gaps have plain meanings:

- $u - \Phi$: how far the value is above immediate exercise;
- $Au - b$: how far the discrete equation inequality is from binding.

Complementarity says that each interior node should be in one of two modes. In a continuation-like node, $u > \Phi$ and the equation should be close to equality. In an exercise-like node, $u = \Phi$ and the equation inequality can be positive. The product of the two gaps should therefore be close to zero.

4 Why Diagnostics Come After Put and Call Validation

Ticket 08 comes after Tickets 05, 06, and 07 because diagnostics need solver outputs to inspect. Those earlier tickets established representative American put and call cases:

- American put: early exercise can matter and the value should dominate payoff and European put values.
- No-dividend American call: the American value should approximately match the European call when $q = 0$.
- Dividend-paying American call: positive dividend yield can create meaningful American-minus-European value.

Ticket 08 does not repeat those price-validation arguments. It checks whether the American solver outputs also satisfy the internal LCP conditions expected from the CN/PSOR formulation.

5 LCP Diagnostic Formulas

For each time step and each interior spot node, Ticket 08 computes:

$$\text{value gap} = u - \Phi, \quad (3)$$

$$\text{equation gap} = Au - b, \quad (4)$$

$$\text{obstacle violation} = \max(\Phi - u, 0), \quad (5)$$

$$\text{equation violation} = \max(b - Au, 0), \quad (6)$$

$$\text{complementarity product} = (u - \Phi)(Au - b). \quad (7)$$

Quantity	Beginner meaning
Value gap	Positive means the option value is above immediate exercise. Zero means the value is at the payoff obstacle.
Equation gap	Positive means the equation inequality has slack. Near zero means the continuation equation is binding.
Obstacle violation	Any positive value means the computed option value fell below payoff.

Quantity	Beginner meaning
Equation violation	Any positive value means the reconstructed matrix inequality $Au \geq b$ was violated.
Complementarity product	Should be close to zero. A large value means the node is neither cleanly continuation-like nor exercise-like under the LCP conditions.

6 Diagnostics Versus Boundary Extraction

Ticket 08 deliberately avoids boundary extraction. It computes value gaps and diagnostic node counts, but it does not search for a transition location, does not interpolate an exercise boundary, and does not label any figure with a boundary. Terms such as exercise-like and continuation-like are local diagnostic classifications only.

This distinction matters. Diagnostics answer: “does the computed value satisfy the LCP conditions?” Boundary extraction asks: “where is the transition curve?” The second question requires a separate algorithm and belongs to a later ticket.

7 Implementation Design Decisions

Decision	Reason
Create <code>src/american_risk_surfaces/diagnostics/lcp.py</code>	Diagnostics are separate from the solver core, keeping pricing logic and validation logic independent.
Use interior nodes only	Boundary nodes are imposed values, not unknowns solved by the LCP. Including them would pollute the residual.
Reconstruct the Ticket 04 matrix form	The diagnostic must check the same A and b that the solver used, including new-time boundary adjustments.
Report maxima and minima	Maximum residuals reveal localized failures that averages can hide.
Use representative put and dividend-call cases	The ticket checks diagnostics on two important American regimes without becoming a full grid sensitivity study.
Use three simple figures	Obstacle violation, equation violation, and complementarity product each answer a different diagnostic question.

8 Code-Level Function Explanations

Function or class	Role
<code>LCPGapArrays</code>	Stores the five interior-node gap arrays for one time step.
<code>ReconstructedLCPStep</code>	Stores u , Φ , Au , b , and gaps for one time step.
<code>StepDiagnosticRow</code>	Stores scalar metrics for one time step, including PSOR metadata.

Function or class	Role
LCPDiagnosticSummary	Stores whole-result maxima, tolerances, convergence status, and diagnostic status.
compute_lcp_gap_arrays	Computes value gap, equation gap, violations, and complementarity product from arrays.
reconstruct_lcp_step	Rebuilds the interior-node LCP for one stored American CN/PSOR step.
summarize_lcp_step	Converts reconstructed arrays into scalar diagnostics and node counts.
diagnose_lcp_result	Runs reconstruction and summary over every PSOR time step.
run_diagnostic_cases	Runs the representative experiment cases and prepares CSV/figure metadata.

9 Diagnostic Cases and Parameters

Case	Option	K	T	r	q	σ	M	N
Put medium	put	1.0	1.0	0.05	0.02	0.20	120	120
Dividend call	call	1.0	1.0	0.05	0.08	0.20	120	120

Both cases use $S_{\max} = 4.0$. The purpose is diagnostic correctness and interpretation, not full grid or domain sensitivity.

10 Diagnostic Result Tables and Interpretation

10.1 Summary Diagnostics

Case	Status	Max obstacle	Max equation	Max comp. product	Max final update	Max iter.
Put	PASS	0	1.5566×10^{-9}	6.6483×10^{-11}	9.9971×10^{-9}	10
Call	PASS	0	2.1442×10^{-9}	2.3796×10^{-11}	9.4309×10^{-9}	10

Both cases pass the Ticket 08 diagnostic thresholds. The maximum obstacle violation is zero in both cases. Equation violations are about 10^{-9} , and complementarity products are about 10^{-11} . These numbers are below the diagnostic thresholds of 10^{-6} for equation violation and complementarity product.

10.2 Node Classification Summary

Case	Max exercise-like nodes	Max continuation-like nodes	Max ambiguous nodes
american_put_medium	113	51	0
dividend_call_medium	114	25	0

The node classifications are diagnostic labels only. They show that nodes can be separated into exercise-like and continuation-like categories under the stated tolerances. They do not locate a free boundary.

10.3 Selected By-Step Rows

Case	Step	τ	Iter.	Final update	Max eq. viol.	Max comp.	Ex-like	Cont-like
put	1	0.00833	10	3.3486×10^{-9}	3.3576×10^{-11}	9.5176×10^{-13}	113	6
put	60	0.50000	7	1.7795×10^{-9}	1.6431×10^{-10}	6.3703×10^{-12}	86	33
put	120	1.00000	6	8.8380×10^{-9}	1.0091×10^{-9}	6.4760×10^{-11}	68	51
call	1	0.00833	10	5.8695×10^{-9}	1.2219×10^{-9}	1.2466×10^{-12}	114	5
call	60	0.50000	7	4.5496×10^{-9}	4.2995×10^{-10}	1.2760×10^{-11}	98	21
call	120	1.00000	7	3.8139×10^{-9}	1.1096×10^{-9}	1.3338×10^{-11}	94	25

The selected rows show that diagnostics remain small early, halfway through, and at final maturity horizon. The full by-step CSV contains all 240 rows.

11 Figure Interpretation

Figure 1 plots maximum obstacle violation by time step. Both cases remain at zero on the generated grid, showing that the stored American values do not fall below payoff.

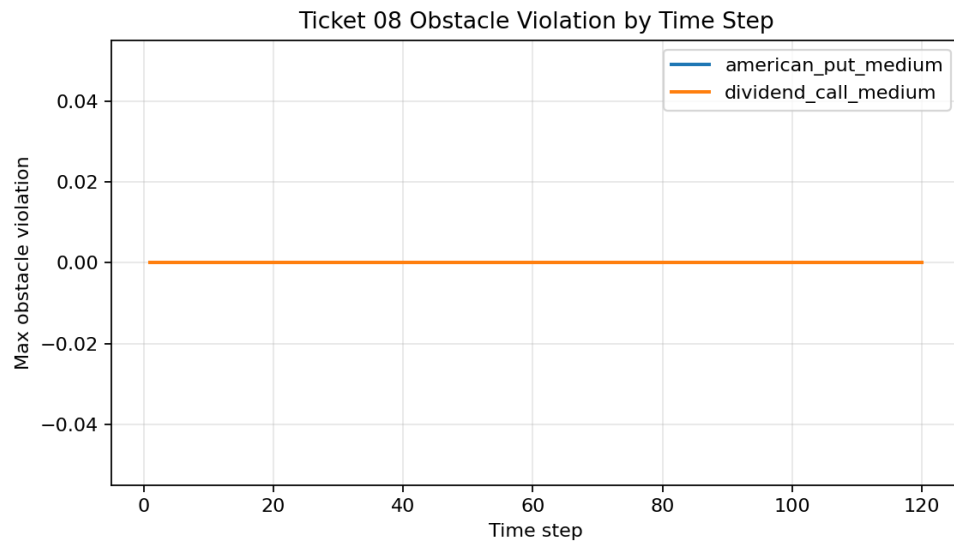


Figure 1: Ticket 08 maximum obstacle violation by time step.

Figure 2 plots maximum equation violation by time step. The values remain near the PSOR tolerance scale and well below the Ticket 08 review threshold.

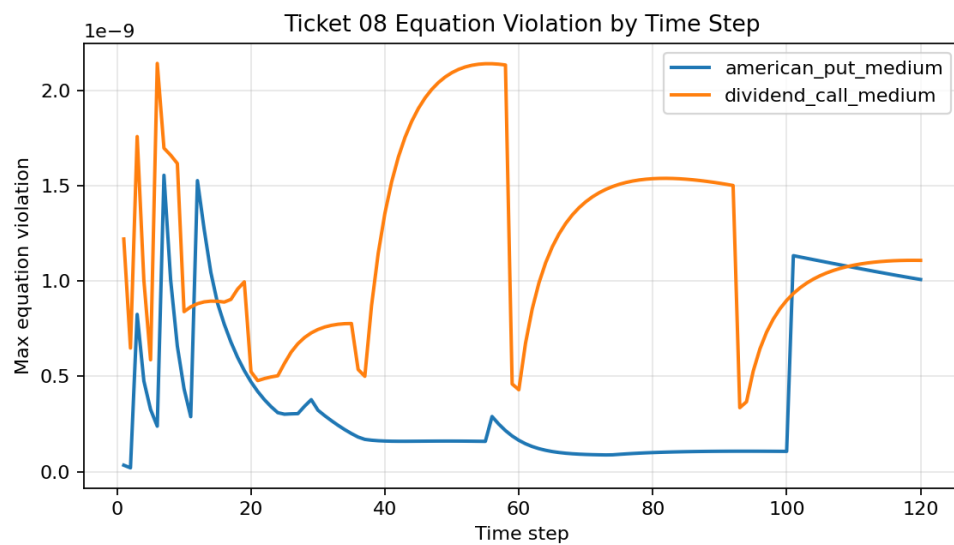


Figure 2: Ticket 08 maximum equation violation by time step.

Figure 3 plots maximum absolute complementarity product by time step. Small values indicate that nodes are close to the LCP alternatives: either continuation-like or exercise-like.

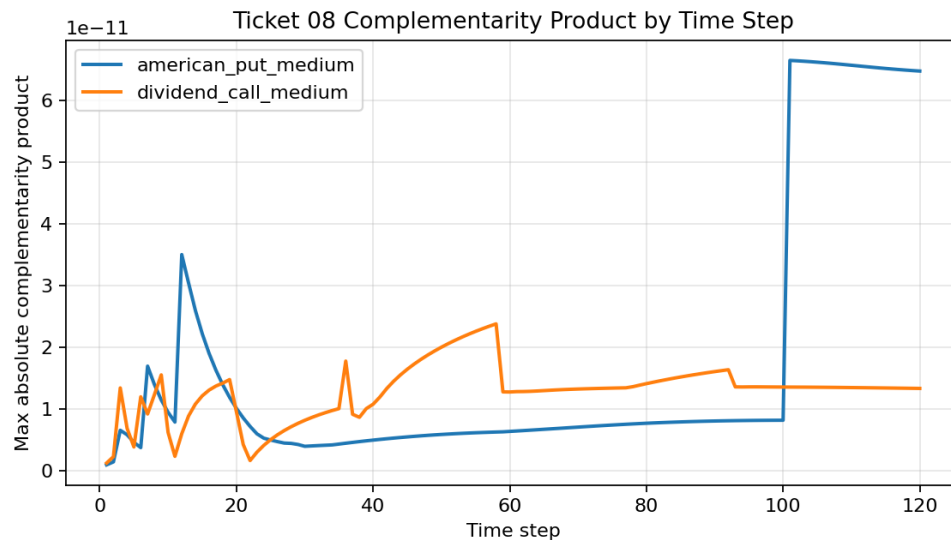


Figure 3: Ticket 08 maximum absolute complementarity product by time step.

12 Testing Strategy

Test category	What it proves	What it does not prove
Manual gap arrays	Basic formulas and signs are correct on small arrays.	Full solver diagnostics.
Toy LCP classification	Exercise-like and continuation-like counts behave as intended.	Boundary location.
Interior reconstruction	Boundary nodes are excluded from residual arrays.	Boundary values are economically perfect.
American put diagnostics	Representative put residuals are finite and small.	Full put grid sensitivity.
Dividend-call diagnostics	Representative positive-dividend call residuals are finite and small.	Full dividend sensitivity.
CSV artifact tests	Experiment outputs have stable schemas.	That later reports need no extra columns.
Figure tests	Meaningful diagnostic figures are created when matplotlib is available.	That plots alone prove correctness.
Scope tests	Ticket 08 does not introduce forbidden APIs/files.	Future tickets that intentionally add those features.

13 Test Results and Interpretation

The final unittest command was:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3 -m
unittest discover -s tests
```

Observed result:

```
Ran 87 tests
```

```
OK
```

The Ticket 08 experiment command was:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3
experiments/05_obstacle_complementarity_diagnostics.py
```

It wrote two summary rows, 240 by-step rows, and three PNG figures.

14 Implementation Pitfalls and Debugging Notes

Recorded item or pitfall	Symptom	Fix or guard	Beginner lesson
Expected TDD red run	The new tests failed because <code>diagnostics/lcp.py</code> and the Ticket 08 experiment script did not exist yet.	Add the diagnostics package and experiment script.	A red test confirms the tests target new behavior.
Malformed targeted unittest command	<code>tests.test_lcp_diagnostics</code> could not import because <code>tests/</code> is not a package.	Use unittest discovery with <code>-s tests</code> and <code>-k</code> .	Match test commands to the repository's import layout.
Wrong equation-gap sign	Reporting $b - Au$ as the gap would reverse the LCP inequality.	Define equation gap as $Au - b$ and equation violation as $\max(b - Au, 0)$.	Sign conventions must be written down before interpreting residuals.
Including boundary nodes	Boundary values are imposed, not solved by the interior LCP.	Slice all diagnostic arrays with <code>1:-1</code> .	Residuals should match the unknowns solved by PSOR.
Overinterpreting classifications	Exercise-like counts can look like an exercise region.	State that counts are diagnostic labels only.	Diagnostics are not boundary extraction.
Ignoring PSOR convergence	Small gaps are less meaningful if the LCP solve did not converge.	Report PSOR iterations, final update, and convergence status.	Solver metadata is part of diagnostic credibility.

15 Limitations

Not included in Ticket 08	Reason
New pricing algorithm	The ticket diagnoses existing American CN/PSOR outputs only.
Free-boundary extraction	Boundary location belongs to a later ticket.
Continuation-premium boundary module	Ticket 08 uses local gaps only, not transition extraction.
Greeks	Delta and Gamma diagnostics are later work.
Stress maps, datasets, neural networks	These require a more complete validated solver pipeline.
Full grid/domain sensitivity	The two medium-grid cases are representative diagnostics, not a convergence study.
Formal proof of complementarity convergence	The report gives numerical diagnostic evidence only.

16 Lessons Learned / Study Notes

The first lesson is that obstacle validity is necessary but not sufficient. A value can stay above payoff while still failing the equation side of the LCP.

The second lesson is that complementarity is a local either-or condition. At each interior node, either the option behaves like a continuation PDE point, or it sits on the payoff obstacle.

The third lesson is that boundary nodes should not be included in LCP residuals. They are imposed by boundary conditions and are not unknowns solved by PSOR.

The fourth lesson is that diagnostic plots should answer diagnostic questions. Ticket 08 uses figures for obstacle violation, equation violation, and complementarity product; it avoids figures that could be mistaken for boundary extraction.

17 Link to the Next Ticket

Ticket 08 provides the reusable diagnostic layer needed before boundary extraction. The next ticket can use value and complementarity evidence with more confidence, but it must still define its own boundary extraction rules, thresholds, interpolation choices, and reporting language.

18 Reproducibility Record

Item	Record
Diagnostics package	<code>src/american_risk_surfaces/diagnostics/lcp.py</code> .
Experiment	<code>experiments/05_obstacle_complementarity_diagnostics.py</code> .
Tests	<code>tests/test_lcp_diagnostics.py</code> .
Summary CSV	<code>results/01_solver_validation/tables/ticket_08_lcp_diagnostics_summary.csv</code> .
By-step CSV	<code>results/01_solver_validation/tables/ticket_08_lcp_diagnostics_by_step.csv</code> .

Item	Record
Figures	results/01_solver_validation/figures/ticket_08_lcp_obstacle_violation_by_step.png; results/01_solver_validation/figures/ticket_08_lcp_equation_violation_by_step.png; results/01_solver_validation/figures/ticket_08_lcp_complementarity_by_step.png.
Report source	reports/03_solver/tickets/ticket_08_obstacle_complementarity_diagnostics.tex.
Report PDF	reports/03_solver/tickets/ticket_08_obstacle_complementarity_diagnostics.pdf.

Commands used:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3
experiments/05_obstacle_complementarity_diagnostics.py
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3 -m
unittest discover -s tests
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src python3 -m compileall
-q src tests experiments
```

References

- Michael J. Brennan and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449, 1977. doi: 10.2307/2326779.
- Richard W. Cottle, Jong-Shi Pang, and Richard E. Stone. *The Linear Complementarity Problem*. Society for Industrial and Applied Mathematics, 2009. ISBN 9780898716863. doi: 10.1137/1.9780898719000.